

# Kyu Hyun Lee

kyuhyun9056@gmail.com | Google Scholar | Website

## ACADEMIC APPOINTMENTS

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**Postdoctoral Scholar, University of California, San Francisco** 2020–present  
Advisor: Loren M. Frank

## EDUCATION

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**California Institute of Technology** 2013–2020  
Ph.D., Biochemistry and Molecular Biophysics option  
Advisor: Markus Meister

**Pomona College** 2007–2011  
B.A., Chemistry (minor: Mathematics), *magna cum laude*

## FELLOWSHIPS AND GRANTS

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**NIH Pathway to Independence Award (K99/R00)** 2025–present  
National Eye Institute (NEI); Principal Investigator; K99EY036953

**Helen Hay Whitney Foundation Postdoctoral Fellowship** 2022–2025  
Tianqiao and Chrissy Chen Institute Graduate Innovator Grant 2018  
Caltech-Amgen Graduate Fellowship 2014–2015

## SELECTED RESEARCH PUBLICATIONS AND MANUSCRIPTS

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1. **Lee, K.**, Adenekan, P., Gao, F., Zhou, J., Hernandez, J., Yorita, A., Haque, R., Kay, K., Scanziani, M., and Frank, L.M. Visual cues modulate abstract task-state representations in primary visual cortex. *Manuscript in preparation*.
2. **Lee, K.\***, Denovellis, E.\*, Ly, R., Magland, J., Soules, J., Comrie, A.E., Gramling, D.P., Guidera, J.A., Nevers, R., Adenekan, P., Brozdowski, C., Bray, S., Monroe, E., Bak, J., Coulter, M., Sun, X., Tritt, A., Rübel, O., Nguyen, T., Yatsenko, D., Chu, J., Kemere, C., Garcia, S., Buccino, A., and Frank, L.M. (2025). Spyglass: a data analysis framework for reproducible and shareable neuroscience research. *eLife*, 14: RP108089.
3. **Lee, K.\***, Ni, Y.\*, Colonell, J., Karsh, B., Putzeys, J., Pachitariu, M., Harris, T.D., and Meister, M. (2021). Electrode pooling can boost the yield of extracellular recordings with switchable silicon probes. *Nature Communications*, 12: 5245.
4. **Lee, K.**, Tran, A., Turan, Z., and Meister, M. (2020). The sifting of visual information in the superior colliculus. *eLife*, 9: e50678.

\* Equal contribution.

## BOOK

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Meister, M., **Lee, K.**, and Portugues, R. (2025). *Mathematics in Biology*. MIT Press.

## ADDITIONAL PEER-REVIEWED PUBLICATIONS

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5. Guidera, J.A., Gramling, D.P., Comrie, A.E., Joshi, A., Tseng, S.Y., Denovellis, E.L., Smyth, C.N., **Lee, K.H.**, Zhou, J., Thompson, P., Hernandez, J., Yorita, A., Haque, R., Kirst, C., and Frank, L.M. (2026). Regional specialization in prefrontal cortex manifests in the reliability of task progression codes. *Neuron*, 114(6): 1152–1161.e8.
6. **Lee, K.H.**, Humbarger, S., Bahnvadia, R., Sazinsky, M.H., and Crane, E.J. III. (2014). Characterization of the mechanism of the NADH-dependent polysulfide reductase (Npsr) from *Shewanella loihica* PV-4: Formation of a productive NADH-enzyme complex and its role in the general mechanism of NADH and FAD-dependent enzymes. *Biochimica et Biophysica Acta - Proteins and Proteomics*, 1844(9): 1708–1717.
7. Warner, M.D., Lukose, V., **Lee, K.H.**, Lopez, K., Sazinsky, M.H., and Crane, E.J. III. (2011). Characterization of an NADH-dependent persulfide reductase from *Shewanella loihica* PV-4: Implications for the mechanism of sulfur respiration via FAD-dependent enzymes. *Biochemistry*, 50(2): 194–206.

## PEER-REVIEWED CONFERENCE PROCEEDING

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Hermiz, J., Joseph, E., **Lee, K.H.**, Baldacci, I.A., Chung, J.E., Frank, L.M., Bouchard, K.E., and Denes, P. (2021). The impact of reducing signal acquisition specifications on neuronal spike sorting. *43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 5914–5918.

## PATENT

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Meister, M., **Lee, K.**, and Ni, Y. (2024). Methods and systems for electrode pooling. U.S. Patent No. 12,097,033.

## RESEARCH MENTORING

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**Denisse Morales-Rodriguez**, Graduate Student, UCSF (2024–present) — Optogenetic manipulation of mPFC during sleep.

**Scott Gao**, Research Associate, UCSF (2024–present) — Hippocampus–visual cortex interactions.

**Philip Adenekan**, Research Associate, UCSF (2022–2024) — Hippocampus–visual cortex interactions.

**Yangsun Hwang**, Rotation Student, UCSF (2023) — Chronic tracking of neurons recorded with polymer probes.

**Roshanak Gonzalez**, Rotation Student, UCSF (2022) — Testing an optoelectronic neural probe.

**Daryl Lawrence**, Rotation Student, UCSF (2021) — Denoising signals from a neural-recording amplifier chip.

## INVITED TALKS AND PRESENTATIONS

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| Navigating Educational Trajectories in Neuroscience (NIH R25 Program), University of Oregon | 2024, 2025 |
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| Dynamic Brain Clusters Databinge, University of British Columbia Center for Brain Health | 2024 |
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| BRAIN Initiative Meeting, “Reproducible Workflows for Collaborative Neuroscience” symposium, NIH | 2023 |
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| Spike Sorting Workshop, Janelia Research Campus | 2018 |
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| Junior Scientist Workshop on Neural Circuits and Behavior, Janelia Research Campus | 2017 |
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## TEACHING EXPERIENCE

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### Division of Biology and Biological Engineering, California Institute of Technology

|                                         |            |
|-----------------------------------------|------------|
| Principles of Neuroscience (Fall terms) | 2018, 2019 |
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| Tools of Neurobiology (Fall terms) | 2016, 2017 |
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| Mathematics in Biology (Fall terms) | 2015, 2016 |
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| Brain Circuits (Winter terms) | 2014, 2015 |
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## HONORS AND AWARDS

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| Phi Beta Kappa | 2011 |
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| Robert Rowan III Memorial Award | 2011 |
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